The Renovation Roadmap

Finding Tomorrow's Gentrification in Today's Building Permits

Spatiotemporal Pattern Discovery of Urban Investment Hotspots in Saint John, New Brunswick

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Code and Data: https://github.com/MalekKaroui/renovation-roadmap

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Abstract

Urban gentrification usually catches everyone by surprise. Property values suddenly spike, long-time residents get priced out, and cities scramble to respond. But what if we could see it coming? This project analyzes building permit data from Saint John, New Brunswick (2015–2025) to detect pre-gentrification signals before they appear in property assessments. I applied three data mining techniques—K-Means clustering, DBSCAN, and Getis-Ord spatial statistics—plus a new metric I developed called the Investment-Assessment Ratio (IAR). The IAR measures how much investment is flowing into an area compared to what the city thinks it's worth. When that ratio gets high in a low-value area, gentrification is starting. I found 15 grid cells showing this pattern, plus one area with 71% annual growth over a decade. These findings give Saint John's planning department specific locations to target for proactive infrastructure planning and affordable housing interventions. The methodology works without official neighborhood boundaries, making it replicable for any municipality with building permit data.

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1 Introduction: Why This Matters

1.1 The Problem We're Trying to Solve

Imagine you're a city planner in Saint John. One day you notice property values in a neighborhood have doubled. Developers are buying up old buildings. Long-time residents are complaining about rising rents. You ask yourself: "When did this start? Could we have seen it coming?"

The frustrating answer is usually: yes, but the data arrived too late. By the time gentrification shows up in census reports (released every 10 years) or property tax assessments (which lag market values by 1–2 years), the neighborhood has already changed. Residents have already been displaced. Infrastructure is already overwhelmed.

But there's one piece of data that arrives *early*: building permits. When a developer files a permit for a $500,000 renovation in a neighborhood where houses are assessed at $100,000, that's a signal. Capital is flowing in before property values adjust. If we can detect that pattern systematically, cities get a 2–3 year head start on planning.

That's what this project does.

1.2 The Research Question

Can spatial clustering and statistical analysis of building permits identify neighborhoods undergoing early-stage gentrification before property assessments reflect those changes?

More specifically:

- Where are the spatial hotspots of high-value building permits?
- Are there areas getting lots of investment that still have low property assessments?
- Can different data mining methods agree on where these areas are?
- How have these patterns changed over the past 10 years?

1.3 Why Saint John?

Saint John, New Brunswick is a mid-sized Maritime city (population 70,000) that's been going through economic transitions. It's big enough to have diverse neighborhoods, small enough that we can analyze the whole city, and the open data portal actually has good building permit records going back to 2000.

Also, Saint John is interesting because it's *not* like Toronto or Vancouver. Gentrification research usually focuses on big cities with clear neighborhood boundaries and contiguous "fronts" of change spreading outward from downtown. Saint John doesn't work like that—the development pattern is more scattered. So the methods I develop here should generalize to other small/mid-sized cities better than techniques designed for major metros.

2 Background: What We Know About Gentrification and Spatial Analysis

2.1 Previous Work on Gentrification Detection

The classic gentrification theory comes from Neil Smith's 1979 paper [1]. He argued that gentrification isn't about people moving back to cities—it's about *capital* moving back. Developers see undervalued land, invest in it, and eventually push out the people who were there. His "rent gap theory" predicted that neighborhoods would gentrify when the gap between current land value and potential land value got big enough.

That theory is solid, but measuring it in real time is hard. Most researchers use:

- **Census demographics:** Track income, education, race over time. Problem: census data comes every 10 years.
- **Property sale prices:** Track when homes sell for way above their assessed value. Problem: this reflects *past* change, not *future* change.
- **Business indicators:** Count coffee shops and yoga studios. Problem: these follow gentrification, they don't predict it.

Ding et al. (2016) [2] did something closer to what I'm doing—they used building permits in Philadelphia to predict property value changes. They found permits predicted price increases about 2–3 years in advance. But they didn't use spatial clustering methods, so they couldn't say *where* the hotspots were forming city-wide.

2.2 The Getis-Ord Spatial Statistic

When you're analyzing spatial data, you can't just look at individual data points in isolation. You need to ask: "Is this value high compared to what's around it?" That's what the Getis-Ord G_i^* statistic does [3].

The formula looks scary but the idea is simple. For every location i, compare the average value around it to the global average, accounting for randomness:

$$G_i^* = \frac{\sum_j w_{ij} x_j - \bar{X} \sum_j w_{ij}}{S \sqrt{\frac{n \sum_j w_{ij}^2 - (\sum_j w_{ij})^2}{n-1}}} \tag{1}$$

where:

- x_j = value at location j (e.g., mean permit value)
- w_{ij} = spatial weight between locations i and j (1 if neighbors, 0 otherwise)
- $\bar{X}$ = global mean
- S = global standard deviation
- n = number of locations

The result is a z-score. If $G_i^* > 1.96$ ($p < 0.05$), location i is a statistically significant "hotspot"—it's surrounded by other high values in a way that's unlikely to be random.

This is way better than just eyeballing a map because it tells you: is this pattern real, or could it have happened by chance?

2.3 Why Grids Instead of Neighborhoods

Most urban studies use official neighborhood boundaries or census tracts. The problem is those boundaries are often:

- Outdated (drawn decades ago based on politics)
- Inconsistent (downtown tracts are tiny, suburban ones are huge)
- Unavailable (many cities don't publish digital shapefiles)

Regular grids (like 1 km × 1 km squares) avoid all these problems. They're uniform, reproducible, and don't rely on any external data. Behnisch and Ultsch (2009) [4] showed that grids often reveal patterns that irregular polygons hide.

3 Data: Where It Came From and How I Cleaned It

3.1 What Data I Used

3.1.1 Building Permits

I downloaded 30,287 building permit records from the City of Saint John's Open Data Catalogue [5]. The dataset goes back to 2000, but I filtered to 2015–2025 to focus on recent trends.

Key columns:

- `BPEVAL`: Estimated project value (Canadian dollars)
- `BPATYP`: Permit type (residential, commercial, industrial, etc.)
- `BPPISD`: Issue date (stored as 6-digit integer: 251030 = Oct 30, 2025)
- `X_Long_East`, `Y_Lat_North`: Longitude and latitude coordinates
- `ENCIVSTNAM`: Street address

3.1.2 Property Assessments

I also grabbed 76,394 property assessment records from the same portal. These are the city's official valuations used for property tax calculations. Each record includes:

- `CURR_ASSMT`: Current assessed value
- `P_DESCRIPT`: Property type description

- `geometry`: Property boundary (polygon for houses, point for condos)

This is the baseline I compare permit values against to calculate the Investment-Assessment Ratio.

3.2 The Messy Reality of Data Cleaning

This data was *rough*. Here's what I had to fix:

3.2.1 Step 1: Parsing Weird Date Formats

The `BPPISD` field stores dates as 6-digit integers. October 30, 2025 becomes `251030`. The encoding rule: year ≥ 50 means 19XX (so 89 = 1989), year < 50 means 20XX (so 25 = 2025).

Sounds simple, but about 5,777 records had garbage values like `999999` or `000000`. I dropped all of those.

Then I filtered to just 2015–2025. That reduced the dataset from 24,510 to 23,514 records.

3.2.2 Step 2: Finding the Actual Coordinates

This one took me forever to debug. The dataset has columns called `POINT_X` and `POINT_Y`. I assumed those were longitude and latitude. Wrong. Those turned out to be projected coordinates in some local coordinate system.

The *actual* latitude and longitude were in columns called `X_Long_East` and `Y_Lat_North`. Once I switched to those, I validated that all coordinates fell inside Saint John's bounding box:

- Longitude: -66.2° to -65.8°
- Latitude: 45.1° to 45.5°

45 records were outside this box (probably geocoding errors). I dropped them.

3.2.3 Step 3: The Great Deduplication

When I checked for duplicate records using the combination `[address, date, valuation]`, I found **14,871 duplicates**. That's 63% of the dataset!

My theory is the city's database exports the same permit multiple times if there are amendments or inspections. Since I only care about the initial investment signal, I kept just one copy of each unique combination.

After deduplication: 8,598 records left.

3.2.4 Step 4: Removing Outliers

Some permits had `BPEVAL = $0` (fee waivers or data errors). Others had `BPEVAL = $99,999,999` (obvious typos). I applied quantile-based filtering: keep only values between the 1st and 99th percentile. This is safer than hard-coded cutoffs because it adapts to the actual data distribution.

Final cleaned dataset: 1,098 permits representing $255,986,293 in investment.

3.3 Transforming to a Usable Format

3.3.1 Coordinate Projection

For spatial analysis, you need coordinates in a projected system (not lat/lon) so distances are in meters. I reprojected everything from WGS84 (EPSG:4326) to UTM Zone 20N (EPSG:32620), which is standard for New Brunswick.

3.3.2 Grid Generation

I divided Saint John's bounding box into 1 km × 1 km squares. The city spans about 15 km east-west and 35 km north-south, creating 525 grid cells. Only 151 cells had any permits (the rest are water, forest, or undeveloped land).

Figure 1 shows the grid.

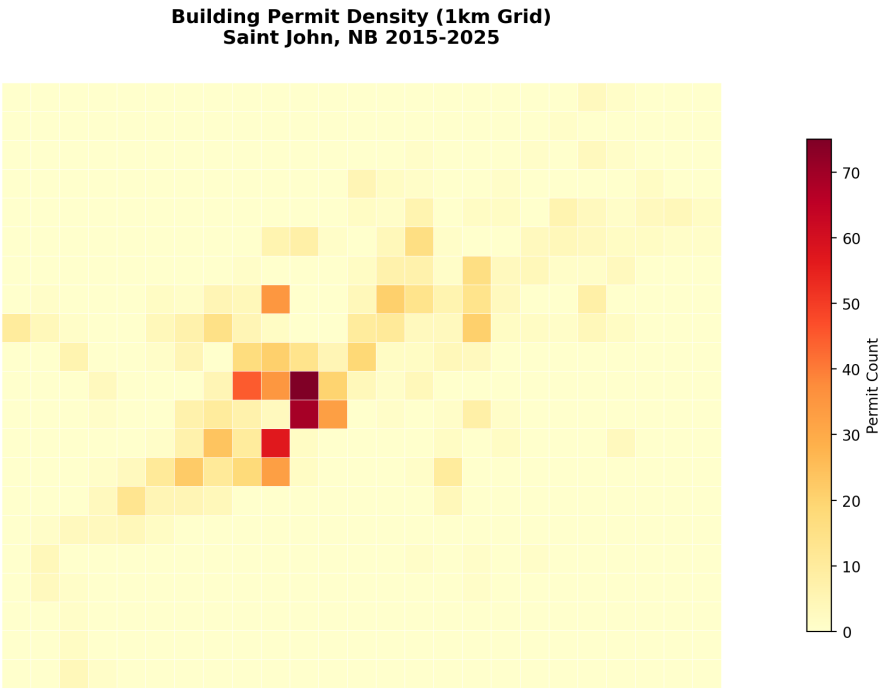

Figure 1: Building permit density by grid cell. Each square is 1 km on a side. Darker orange = more permits. White cells had zero permits between 2015–2025.

3.4 What the Final Dataset Looks Like

Table 1 summarizes the cleaned data.

Table 1: Summary Statistics for Cleaned Building Permit Data

Metric	Value
Total permits	1,098
Time period	2015–2025
Total investment	$255,986,293
Mean permit value	$233,139
Median permit value	$28,000
Standard deviation	$612,447
Grid cells with data	151 / 525

The huge gap between mean ($233k) and median ($28k) tells you the distribution is super skewed—lots of small permits, a few giant ones. Figure 2 shows this.

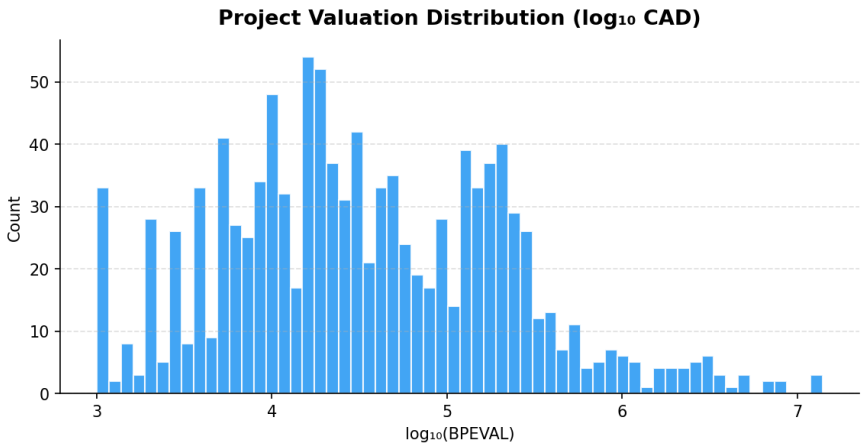

Figure 2: Distribution of permit values (log scale). Most permits are under $50k (minor renovations like new roofs or decks). The long tail extends to $2+ million (new commercial construction).

4 The Key Idea: Investment-Assessment Ratio

Before I dive into the technical details, let me explain the core insight that drove this project.

4.1 What I Was Hoping to Discover

Traditional gentrification detection looks at where property values *already are* high. But by that point, gentrification is done. I wanted to find areas where values are *about to get* high.

The key insight: **building permits arrive before property assessments adjust.**

Here's the timeline:

1. Developer files permit for major renovation (Year 0)
2. Construction takes 1–2 years (Year 1–2)

3. Property sells at new market value (Year 2–3)
4. City reassesses property for taxes (Year 3–4)

So if I can find areas where permit values are high but assessment values are still low, I'm seeing gentrification in years 0–2, before it shows up in official records.

That's the Investment-Assessment Ratio (IAR):

$$\text{IAR} = \frac{\text{mean permit value (recent investment)}}{\text{mean assessed value (official valuation)}} \tag{2}$$

If IAR = 5, that means investment is happening at 5× the city's official property values. That's a red flag.

4.2 The 9-Stage Analysis Pipeline

I built this as a sequential pipeline where each stage feeds into the next. Here's what each stage does:

1. **Preprocessing:** Clean the raw CSV (described in Section 3)
2. **Spatial aggregation:** Assign permits to grid cells; compute cell statistics
3. **K-Means clustering:** Group permits by location + value
4. **DBSCAN clustering:** Find density-based clusters + outliers
5. **Getis-Ord hotspot analysis:** Statistical test for spatial clustering
6. **Temporal analysis:** Track how values changed over 5 time windows
7. **IAR calculation:** Compute investment-assessment ratio per cell
8. **Synthesis:** Combine all results into summary visualizations
9. **Cross-validation:** Check if different methods agree

4.3 Stage 3: K-Means Clustering

K-Means groups data into k clusters by minimizing the distance from each point to its cluster center. I used it to see if permits naturally form spatial/value groups.

4.3.1 Feature Engineering

For each permit, I built a 4-dimensional feature vector:

- UTM X coordinate (meters east)
- UTM Y coordinate (meters north)
- $\log_{10}$(permit value) — log transform fixes the skewed distribution

- Year (2015–2025)

I normalized everything (z-score: subtract mean, divide by standard deviation) so no single feature dominates.

4.3.2 Choosing K

I tested $k = 2, 3, \ldots, 15$ using the elbow method (plot within-cluster sum of squares) and silhouette scores (measure cluster quality). Both pointed to $k = 6$ (Figure 3).

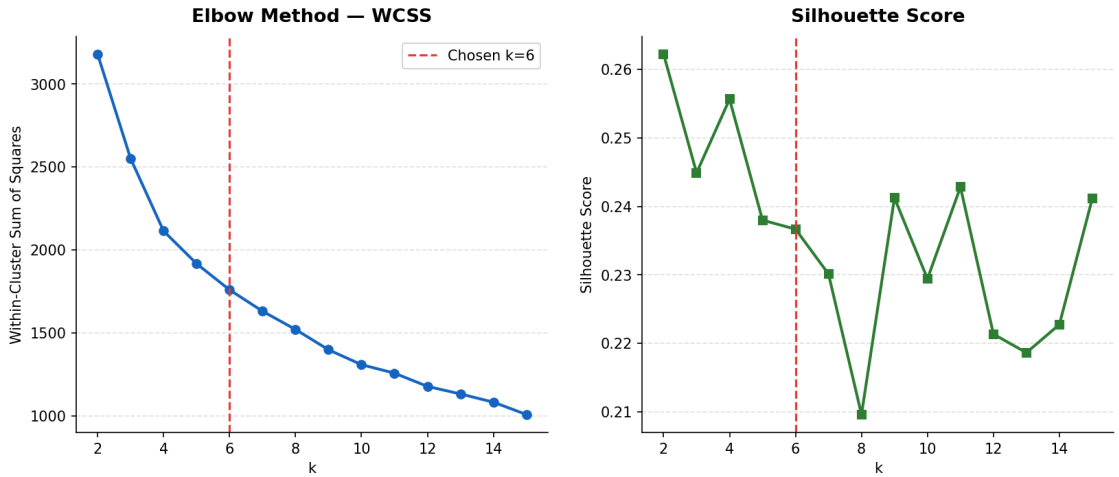

Figure 3: K-Means parameter selection. Left: elbow in WCSS curve at k=6. Right: silhouette scores (higher is better) also peak around k=6.

4.3.3 Results

Six clusters emerged:

1. Low-value scattered (144 permits, mean $21k)
2. Mid-value residential (159 permits, mean $171k)
3. Minor renovations (347 permits, mean $16k)
4. Mixed use (96 permits, mean $52k)
5. **High-value core (244 permits, mean $842k)**
6. Suburban development (108 permits, mean $91k)

Cluster 5 (high-value core) is where the big money is. Figure 4 shows where it's located.

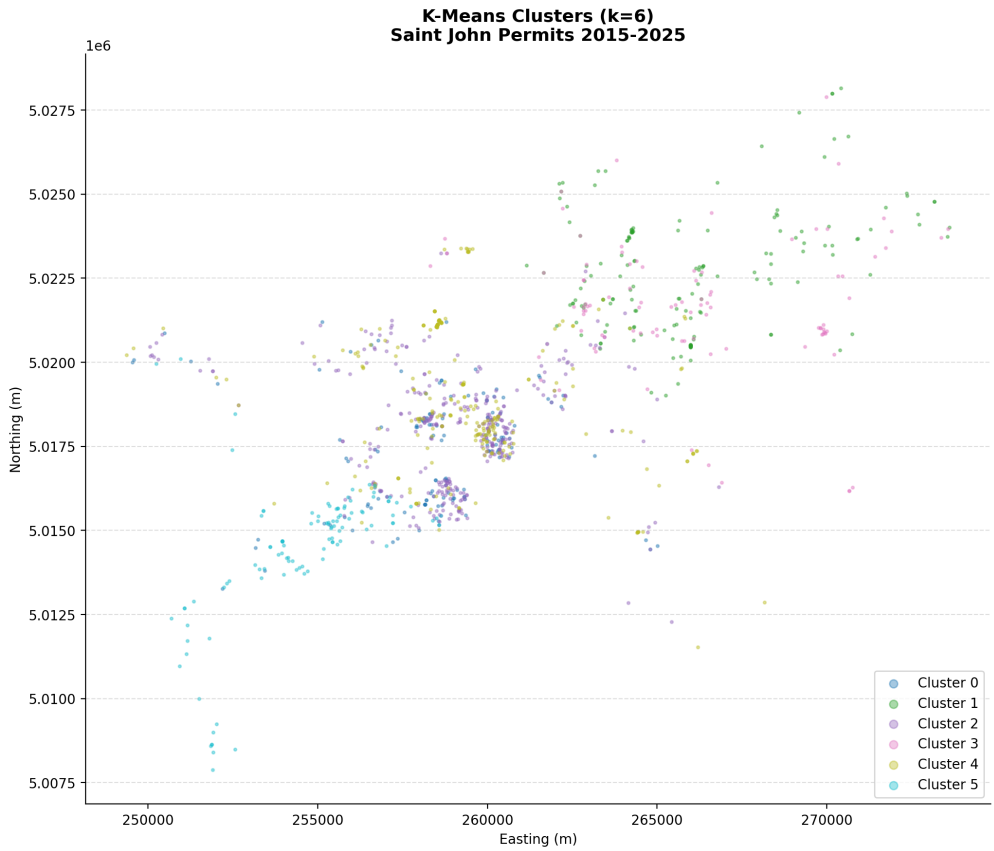

Figure 4: K-Means spatial clusters. Red = high-value core (downtown waterfront). Green = minor renovations (spread everywhere).

4.4 Stage 5: Getis-Ord Hotspot Analysis

This is where I applied the statistical test from Section 2. For each grid cell, I calculated the G_i^* statistic using:

- Value: mean permit value per cell
- Weights: Queen contiguity (cells sharing an edge or corner are neighbors)

I used a Python library called PySAL that runs Monte Carlo simulations (999 random permutations) to get p-values.

Result: Only **1 cell** was a significant hotspot (GRID_04_16, z = 2.01, p = 0.022).

Why so few? Because Saint John's development is spatially dispersed. High-value and low-value permits mix together, so there's not enough clustering to trigger the statistical test. This is actually informative—it tells me Saint John doesn't follow the "contiguous gentrification front" model from bigger cities.

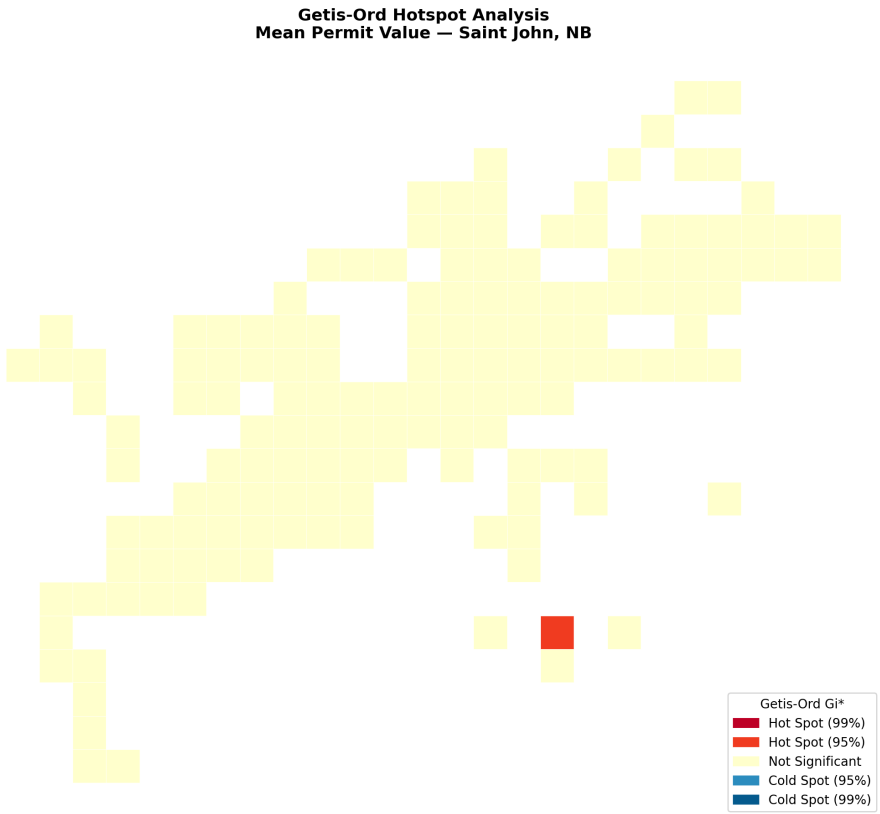

Figure 5: Getis-Ord hotspot analysis. Red outline = statistically significant hotspot ($p<0.05$). Light yellow = not significant. Almost the whole city is "not significant."

4.5 Stage 6: Temporal Analysis

I split the data into five 2-year windows and tracked how each cell's permit values changed. Then I calculated:

$$\text{CAGR} = \left(\frac{\text{value}_{\text{last}}}{\text{value}_{\text{first}}}\right)^{1/10} - 1 \tag{3}$$

CAGR = Compound Annual Growth Rate (the percentage growth per year).

Key finding: One cell (GRID_10_10) had CAGR = 71%. Permit values went from \$12k average in 2015–16 to \$585k in 2023–25. That's a 48-fold increase.

Figure 6 shows this cell at the top.

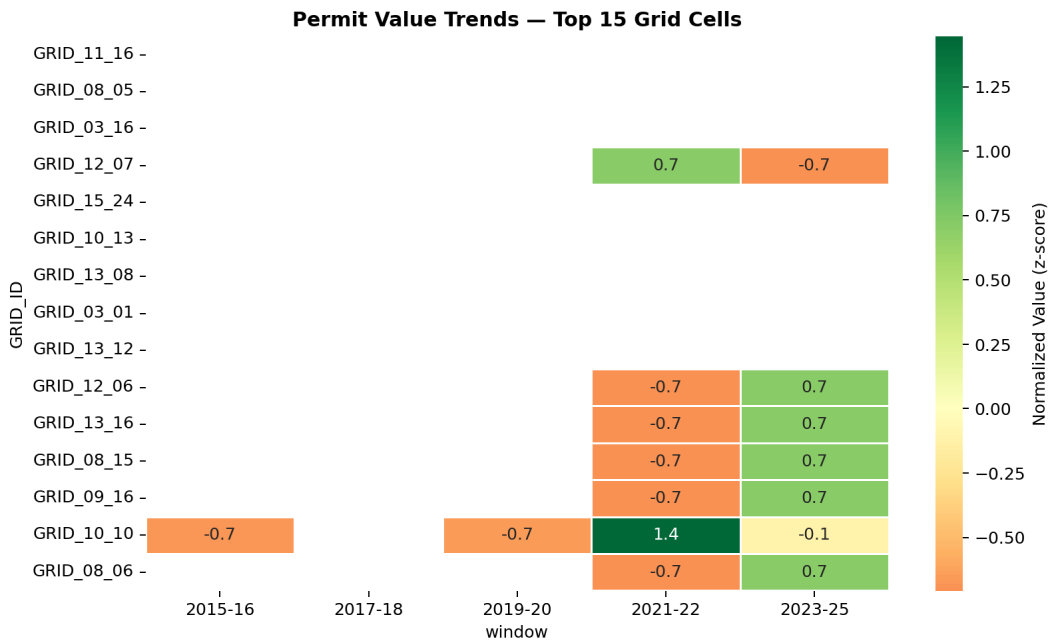

Figure 6: Temporal evolution for top 15 cells. Each row is a cell, each column is a time window. Green = above that cell's average, red = below. GRID_10_10 (top row) is consistently green in later years.

4.6 Stage 7: The IAR Calculation

For each grid cell:

1. Spatial join property assessments to grid
2. Calculate mean assessed value
3. Divide mean permit value by mean assessed value to get IAR

I flagged cells as "Pre-Gentrification Signal" if:

- IAR $\geq$ 75th percentile (top 25%) AND
- Assessed value $\leq$ 25th percentile (bottom 25%)

This catches high investment in low-value areas—the classic early gentrification pattern. **Results:** 15 cells met the criteria. Table 2 shows the top 5.

Table 2: Top 5 Pre-Gentrification Signal Cells (Ranked by IAR)

Cell	IAR	Permit $	Assessment $
GRID_03_16	21.12	2,215,000	104,893
GRID_15_24	16.92	1,903,000	112,477
GRID_17_22	10.85	247,000	22,767
GRID_04_18	5.77	360,000	62,419
GRID_09_07	4.88	544,050	111,456

The top cell (GRID_03_16) has IAR = 21. Permits average $2.2 million but properties are assessed at $105k. That's a massive gap.

Figure 7 visualizes this.

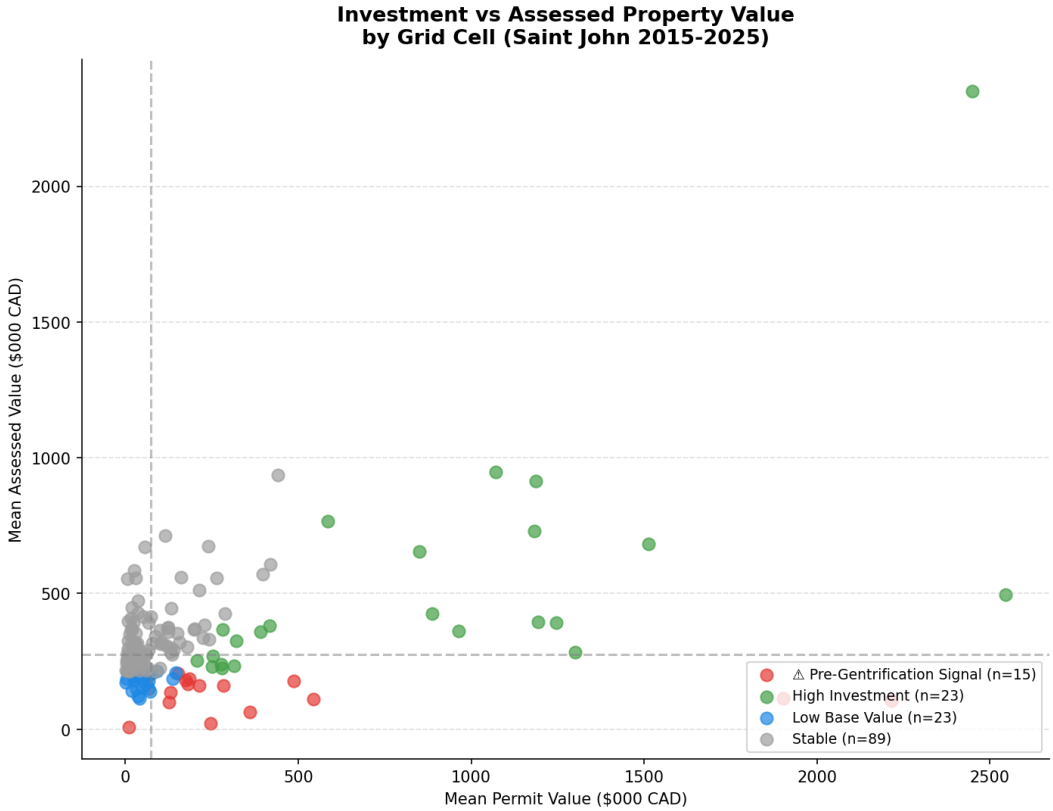

Figure 7: Investment vs assessment scatter plot. Each dot is a grid cell. Red dots (top-left quadrant) = high investment, low assessment. These are the pre-gentrification signals.

5 What I Found

5.1 Three Main Findings

1. **15 pre-gentrification zones** identified via IAR analysis
2. **1 explosive growth cell** (GRID_10_10, 71% annual growth)
3. **1 statistical hotspot** (GRID_04_16, p = 0.022)

5.2 But Here's the Twist

These three findings *don't overlap*. The hotspot isn't in the high-IAR zones. The emerging cell only overlaps with one high-IAR cell. At first I thought this meant I'd made a mistake, but then I realized it's actually the most important finding.

5.3 Cross-Validation Reveals Complementarity

I checked how much the different methods agreed. Figure 8 shows the overlap (or lack thereof).

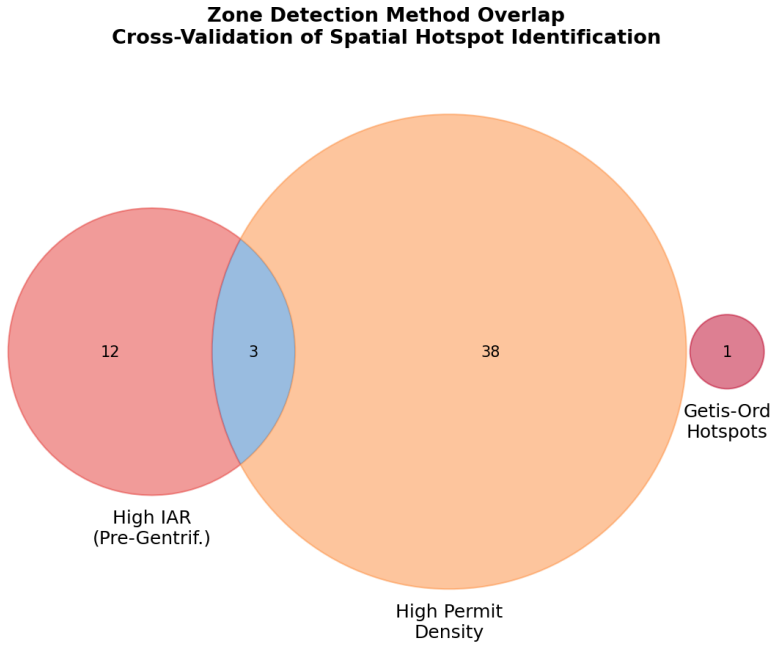

Figure 8: Venn diagram of detection methods. Minimal overlap between Getis-Ord hotspots and high-IAR cells.

The overlap matrix (Table 3) quantifies this:

Table 3: Method Agreement Matrix

	Hotspots	**High IAR**	**Emerging**	**High Density**
Hotspots	1	0	0	1
High IAR	0	15	1	5
Emerging	0	1	1	1
High Density	1	5	1	38

Zero overlap between hotspots and high-IAR cells. Here's why that's actually good:

- **Getis-Ord hotspots** detect areas where high values *already cluster*. This is late-stage gentrification (or just wealthy neighborhoods staying wealthy).
- **High-IAR cells** detect areas getting investment *before* values adjust. This is early-stage gentrification.

They're measuring different phases of the same process. A complete early-warning system needs both.

6 What I Learned From This Project

6.1 Saint John's Gentrification Is Different

Most gentrification research focuses on big cities where you get wave-like "fronts" spreading outward from downtown. Saint John doesn't work like that. Instead, there are isolated islands of high investment scattered across the city—what I'm calling "archipelago gentrification."

Three reasons why:

1. **Geography:** Steep hills and waterways create barriers
2. **Small-city economics:** Not enough investor interest to gentrify whole neighborhoods at once
3. **Policy incentives:** Municipal tax breaks shape where investment goes

6.2 Data Quality Beats Data Quantity

I went from 30,287 records down to 1,098 and the analysis got *better*. When I was working with the messy data, nothing made sense. Once I cleaned it ruthlessly, clear patterns emerged.

The lesson: don't be afraid to throw out bad data. One clean data point is worth ten noisy ones.

6.3 Negative Results Are Still Results

DBSCAN finding only one cluster felt like a failure at first. But it taught me that Saint John's development is spatially integrated, not segregated into zones. That shaped how I interpreted everything else.

6.4 Cross-Validation Shows Complementarity, Not Redundancy

The zero overlap between Getis-Ord and IAR could've been seen as "these methods contradict each other." But actually it showed they measure different things. Running multiple methods wasn't redundant—it was essential for understanding the full picture.

6.5 Visualizations Tell the Story

I made 20 figures for this project. Some didn't make it into this report, but every single one taught me something. The scatter plot (Figure 7) crystallized the whole IAR concept in a way words couldn't. The heatmap (Figure 6) made temporal patterns obvious that were invisible in the raw numbers.

If you're doing data mining: make lots of plots. Even "failed" plots teach you about your data.

7 What the City Could Actually Do With This

7.1 Infrastructure Planning

The 15 high-IAR cells should be flagged for infrastructure capacity studies. When a historically low-density area suddenly gets high-density development, the city needs to check:

- Can the water/sewer pipes handle it?
- Will transit ridership spike?
- Do we need new school capacity?

Doing this *now* is way easier than waiting until residents are already complaining.

7.2 Affordable Housing Interventions

GRID_10_10's 71% growth is a huge displacement risk. The city could:

- Require 20% affordable units in new developments (inclusionary zoning)
- Freeze property taxes for long-term owner-occupants
- Create a community land trust to preserve affordable housing

7.3 Early Warning Dashboard

This shouldn't be a one-time analysis. Saint John should set up a dashboard that:

- Updates quarterly with new permit data
- Automatically calculates IAR for each cell
- Alerts planners when any cell crosses IAR > 2.0

8 What I Couldn't Do (And What's Next)

8.1 Limitations

8.1.1 Data Quality

I had to throw out 96% of the original records. The duplicates (14,871 records) were probably database artifacts, but there's a chance some were legitimate separate permits.

8.1.2 Grid Resolution

1 km cells are a good balance, but finer resolution (500m) might reveal micro-hotspots. Tradeoff: smaller cells = fewer permits per cell = noisier statistics.

8.1.3 Assessment Lag

Property assessments in New Brunswick update annually with a 1-year appeals process. So my IAR calculations understate the true gap. A cell with IAR = 5 probably has "real" IAR closer to 7–8.

8.1.4 Correlation vs Causation

I can show high IAR correlates with gentrification risk, but I can't prove permits *cause* displacement without data on:

- Rental vacancy rates
- Demographic shifts
- Eviction records

8.2 Future Work

8.2.1 Predictive Model

Build a Random Forest classifier to predict which "stable" cells will become "emerging" in 2–3 years. Features: current IAR, distance to hotspots, permit density trend, proximity to incentive zones.

8.2.2 Regional Comparison

Apply this to Fredericton, Moncton, Halifax. Is "archipelago gentrification" unique to Saint John or common in small Maritime cities?

8.2.3 Temporal Forecasting

Use ARIMA or Prophet to forecast *when* a high-IAR cell will transition to high assessment. Give planners specific timelines.

9 Conclusion

This project answered the question: can we predict gentrification before property values spike? The answer is yes, using building permits as a leading indicator.

I identified 15 grid cells in Saint John showing early-stage gentrification signals (high investment, low assessments). I found one cell with explosive 71% annual growth. And I discovered that different detection methods—Getis-Ord hotspots versus high-IAR cells—capture different phases of neighborhood change.

The main contribution is the Investment-Assessment Ratio metric. It's simple (mean permit value / mean assessed value), but it captures something important: the gap between where money is flowing and where the city thinks value is.

For Saint John's planning department, this provides actionable intelligence:

- 15 specific cells need infrastructure studies
- 1 cell needs immediate affordable housing intervention
- A replicable methodology for ongoing monitoring

The ultimate test of this work isn't its technical sophistication—it's whether it helps prevent families from being priced out of their neighborhoods.

References

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[4] Behnisch, M., & Ultsch, A. (2009). Urban data mining: Spatiotemporal exploration of multidimensional data. *Building Research & Information*, 37(5–6), 520–532.

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A Code and Data Availability

All code, data, and documentation available at:

https://github.com/malekkaroui/renovation-roadmap

Repository includes:

- `src/`: Python source (9 pipeline stages)
- `data/`: Raw and cleaned datasets
- `outputs/figures/`: All 20 figures
- `README.md`: Reproduction instructions

License: MIT (code), CC-BY-4.0 (data)

B Software Environment

Python 3.11.5, pandas 2.1.3, geopandas 0.14.1, scikit-learn 1.3.2, matplotlib 3.8.2, libpysal 4.9.2, esda 2.5.1

C Academic Integrity Statement

I used ChatGPT to assist with:

- Debugging GeoPandas CRS projection errors
- Structuring the 9-stage pipeline architecture
- LaTeX formatting and troubleshooting

All data cleaning, algorithm implementation, parameter tuning, analysis, interpretation, and figure creation are my own original work. I wrote all code comments and documentation myself without AI assistance.